Generation as Computation

Generalised Computation in the Set-theoretic Universe and Beyond
(Thesis Proposal)

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Outline

1. Introduction
   - Background: Infinitary Computation
   - Background: Formulas as Programs
   - Overview
2. Generalised Computation Within V
   - Generalised Sequential Algorithms
   - Comparisons with Other Notions
3. Generalised Computation Beyond V
   - Degrees of Small Extensions
   - Local Method Definitions

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Gödel's Informal Proposition

- In an essay Gödel wrote in 1958 (but only published posthumously) [4], he criticised Hilbert's choice of "intuitively clear" notions meant to make up the bedrock of classical mathematics.
- He lamented that "the domain of 'finitary mathematics"' is unnaturally restrictive.
- In the same work, Gödel informally described a notion of infinitary computation built inductively from natural numbers.
- He called this notion *computable functions of finite type*.

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In Classical Computability Theory

- "A program is a formula (of a particular form)" — an oft-repeated slogan.
- An oracle machine can functionally be represented as a sufficiently simple parametrised formula in the language of arithmetic.
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Implicit Conceptions of Infinitary Computation

- α-recursion theory, born from ideas by Takeuti [1], Kreisel and Sacks [2] and some others, in the 1960s.
- E-recursion theory, formulated by Normann [3] in the 1970s.
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In Greater Generality...

- Relative computability is but a form of parametrised definability over some structure.
- Vanilla computability is a special case whereby the parameter in question contains the least amount of information (e.g. using the empty set as parameter).
- From a set-theoretic perspective, computation can thus be thought of as procedural set (even class) generation over a possibly enriched universe of set theory.

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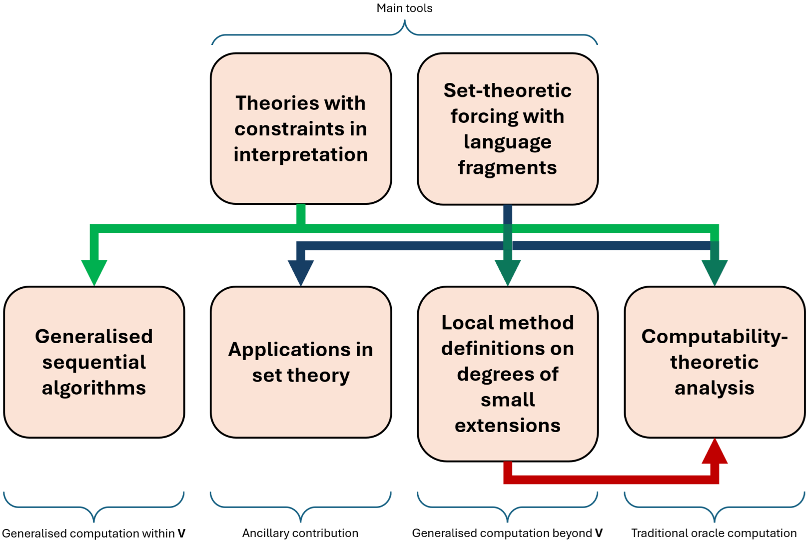
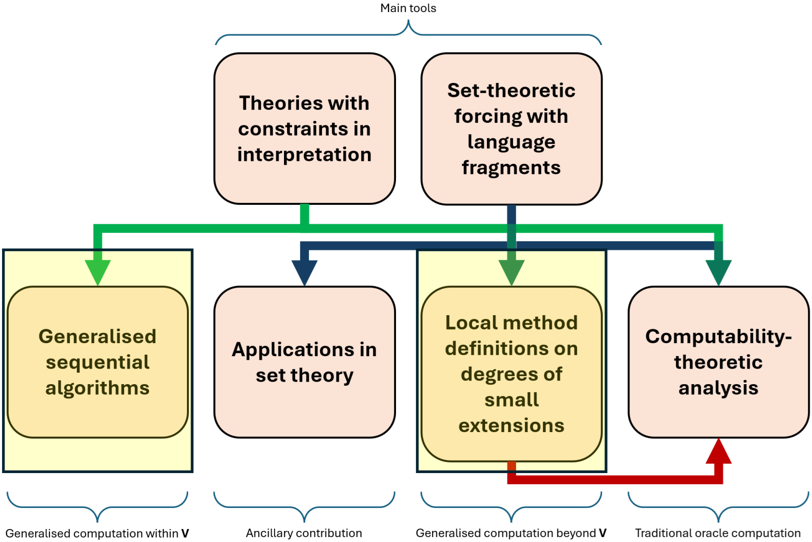
Overview Chart



Today's Focus



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Summary Table

In the following table we compare various relative computability relations when they are restricted to subsets of admissible ordinals α.

Property \ Relation	$\leq_\alpha$	$\preceq_\alpha$	$\leq_\alpha^P$	$\leq_\alpha^{P,s}$
Oracle-analogue?	✓	✓	✗	✗
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Upward persistent?	✗	✓	✓	✓
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The Meta-theory

- What do we mean by "sets outside V"?
- Step out of V and treat V as a countable transitive model of ZFC (henceforth denoted CTM).
- Look at those sets that can be adjoined to V to give a nice CTM.
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Outer Models

- Let U_1 and U_2 be CTMs. U_2 is an outer model of U_1 iff
  - $U_1 \subset U_2$, and
  - $ORD^{U_1} = ORD^{U_2}$.
- The binary relation "being an outer model of" is transitive.

Definition

Let V be a CTM. The *outward multiverse centred at V* is the set

$$\mathbf{M}(V) := \{W : W \text{ is an outer model of } V\}.$$

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Theorem (Jensen)

Given a CTM V, $(\mathbf{M}_S(V), \subset)$ is a cofinal subposet of $(\mathbf{M}(V), \subset)$.

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Let V be a CTM. Then

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Degrees over V

- Specifically, define the binary relation $\leq^S$ on $\mathscr{G}(V)$ by

$$x \leq^S y \iff V[x] \subset V[y].$$

- $\leq^S$ partially orders $\mathscr{G}(V)$, so we can define the equivalence relation $\equiv^S$ the usual way.
- Call $(\mathscr{G} / \equiv^S, \leq^S / \equiv^S)$ the *degrees of small extensions of V* with its standard partial ordering.
- Observe that

$$(\{\text{small extensions of } V\}, \subset) \cong (\mathscr{G} / \equiv^S, \leq^S / \equiv^S),$$

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Drawing Parallels

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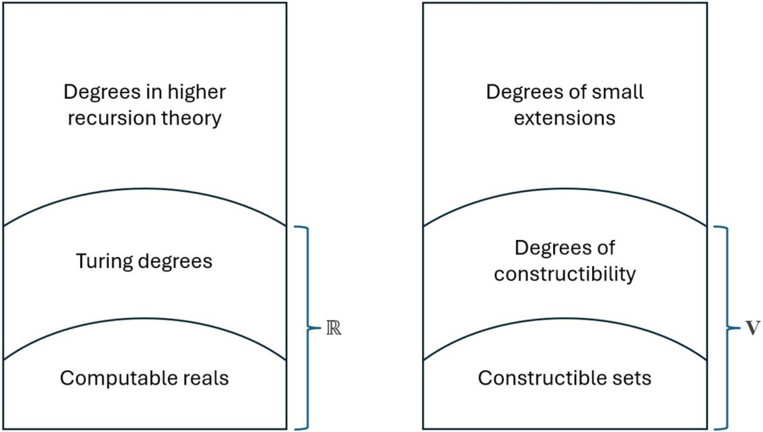

Figure: comparison between conventional notions of relative computability (left) and our generalised notions (right).

Multiverse of Generic Extensions

Definition

Let V be a CTM. The *outward generic multiverse centred at V* is the set

$$\mathbf{M}_F(V) := \{W : W \text{ is a forcing extension of } V\}.$$

The following fact is basic.

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Every generic extension is a small extension. Equivalently,

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Outline

Referring to Small Extensions in V

- Often it is useful to refer to small extensions of V within V.
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Theories with Constraints in Interpretation

- TCIs provide bounds to the interpretation of a theory T over a signature σ.
- A TCI $\mathscr{T}$ expands on T by specifying additional requirements on a possible model $\mathfrak{M}$ of T.
- These requirements include
  - a set upper bound (under $\subset$) to the base set M of $\mathfrak{M}$,
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Consistency of TCIs

Definition (L.)

A TCI is *consistent* iff it has a model in some outer model of V.

By examining a suitable forcing notion in V, we can decide if a TCI in V is consistent.

Proposition

Let $\mathscr{T}$ be a TCI. Then for all sufficiently large cardinals λ,

$$\Vdash_{Col(\omega,\lambda)} \exists \mathfrak{M} \text{ (“}\mathfrak{M} \textit{ is a model of } \mathscr{T}\text{”)} \iff \mathscr{T} \textit{ is consistent.}$$

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Local Method Definitions

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A *local method definition of V* is a non-empty class of TCIs definable in V.

- In the meta-theory, define a function Eval^V from the set of TCIs $\mathscr{T} \in V$ into the set of small extensions of V, such that

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If X and Y are local method definitions of V, $X \leq^M Y$ denotes the statement

"there is a function $F : X \longrightarrow Y$ definable in V such that $\emptyset \neq \mathrm{Eval}^V(F(\mathscr{T})) \subset \mathrm{Eval}^V(\mathscr{T})$ for all consistent $\mathscr{T} \in X$".

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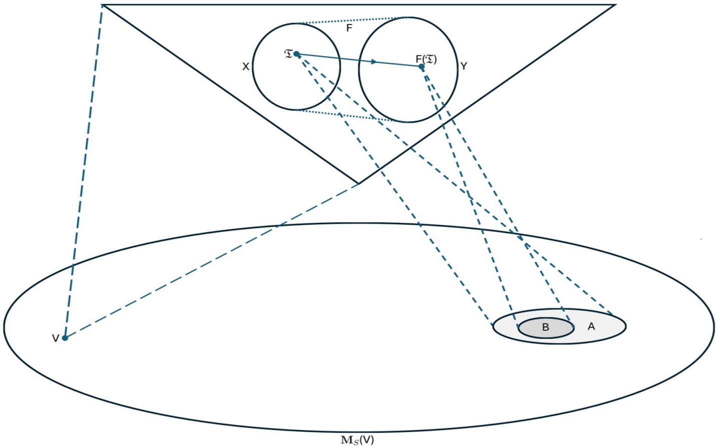

Figure: Visual representation of a function F witnessing $X \leq^M Y$, where X and Y are local method definitions.

The Local Method Hierarchy

- We can also define the complexity of a TCI by just looking at its first-order theory.
- For example, a Σ_n TCI has a first-order theory containing only Σ_n sentences.
- The classes of Σ_n and Π_n TCIs — denoted $\Sigma^{\mathsf{M}}_{\mathsf{n}}$ and $\Pi^{\mathsf{M}}_{\mathsf{n}}$ respectively — then form a hierarchy under the relation $\leq^M$, called the local method hierarchy.

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Let $1 \leq n < \omega$. Then for every $\mathscr{T} \in \Pi^{\mathsf{M}}_{\mathsf{n+1}}$ there is $\mathscr{T}' \in \Sigma^{\mathsf{M}}_{\mathsf{n}}$ such that

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$\Pi^{\mathsf{M}}_{\mathsf{n+1}} \leq^M \Sigma^{\mathsf{M}}_{\mathsf{n}}$ *for all* $1 \leq n < \omega$.

Set Forcing

- There is an obvious way of representing any forcing notion as a Π_2 TCI.
- Thus set forcing, as a technique of accessing small extensions of V, can be represented by a local method definition, denoted Fg.
- We want to see if Fg fits nicely in the local method hierarchy.
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Set Forcing is Σ_1 (is Π_2)

Theorem (L.)

$\mathsf{Fg} \equiv^M \Sigma_1^{\mathsf{M}}$.

Proof Idea.

By adapting our framework for constructing forcing notions to the context of TCIs, we associate with each Π_2 TCI $\mathscr{T}$ a forcing notion $\mathbb{P}(\mathscr{T})$, such that a unique model of $\mathscr{T}$ can be read off every $\mathbb{P}(\mathscr{T})$-generic filter over V.

This allows us to define a class function witnessing $\Pi_2^{\mathsf{M}} \leq^M \mathsf{Fg}$. $\square$

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Further Results

- We can refine the function witnessing $\Pi_2^{\mathrm{M}} \leq^M \mathrm{Fg}$ by iteratively pruning the $\mathbb{P}(\mathscr{T})$s of atoms.
- There are analogues of the previous theorem in V pertaining to certain countable Π_2 TCIs.
  - Essentially, we can define functions F of low complexity taking these TCIs to reals such that every $F(\mathscr{T})$-1-generic real codes a model of $\mathscr{T}$.

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Some Open Questions

(Q1) What structural properties of $(\mathbf{M}_F(L), \subset)$ also hold for $(\mathbf{M}_S(L), \subset)$?

(Q2) For example, is $(\mathbf{M}_S(L), \subset)$ downward directed?

(Q3) Are there $m, n < \omega$ for which $\Sigma^{\mathsf{M}}_{\mathsf{m}} \neq^M \Sigma^{\mathsf{M}}_{\mathsf{n}}$?

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Thank You!